

Numpy Hacker Rank

1. Transpose

We can generate the transposition of an array using the tool `numpy.transpose`. It will not affect the original array, but it will create a new array.

```
import numpy
```

```
my_array = numpy.array([[1,2,3],  
                        [4,5,6]])  
  
print numpy.transpose(my_array)
```

#Output

```
[[1 4]
```

```
[2 5]
```

```
[3 6]]
```

Flatten

The tool *flatten* creates a copy of the input array flattened to one dimension.

```
import numpy
```

```
my_array = numpy.array([[1,2,3],  
                        [4,5,6]])  
  
print my_array.flatten()
```

#Output

```
[1 2 3 4 5 6]
```

Task

You are given a X integer array matrix with space separated elements (= rows and = columns).

Your task is to print the *transpose* and *flatten* results.

Input Format

The first line contains the space separated values of and .

The next lines contains the space separated elements of columns.

Output Format

First, print the *transpose* array and then print the *flatten*.

Sample Input

2 2

1 2

3 4

Sample Output

[[1 3]

[2 4]]

[1 2 3 4]

The screenshot displays the HackerRank interface for the 'Transpose and Flatten' problem. On the left, the problem description states: 'We can generate the transposition of an array using the tool numpy.transpose. It will not affect the original array, but it will create a new array.' Below this, the sample input and output are shown. The sample input is a 2x2 matrix: `[[1,2,3], [4,5,6]]`. The sample output shows the transpose array `[[1 4], [2 5], [3 6]]` and the flattened array `[1 2 3 4 5 6]`.

The main area is a code editor with a Python solution. The code reads input from STDIN, splits it into rows and columns, and then prints the transpose and flattened arrays. The code is as follows:

```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 import numpy as np
3 li=[]
4
5 n,m = map(int,input().strip().split())
6 for _ in range(n):
7     e = input().strip().split(' ')
8     li.append(e)
9
10 arr = np.array(li,dtype='int64')
11 arr.resize(n,m)
12 print(np.transpose(arr))
13 print(arr.flatten())
14
```

At the bottom, there are buttons for 'Upload Code as File', 'Test against custom input', 'Run Code', and 'Submit Code'.

2. Concatenate

Two or more arrays can be concatenated together using the *concatenate* function with a tuple of the arrays to be joined:

```
import numpy
```

```
array_1 = numpy.array([1,2,3])
```

```
array_2 = numpy.array([4,5,6])
```

```
array_3 = numpy.array([7,8,9])
```

```
print numpy.concatenate((array_1, array_2, array_3))
```

#Output

```
[1 2 3 4 5 6 7 8 9]
```

If an array has more than one dimension, it is possible to specify the axis along which multiple arrays are concatenated. By default, it is along the first dimension.

```
import numpy
```

```
array_1 = numpy.array([[1,2,3],[0,0,0]])
```

```
array_2 = numpy.array([[0,0,0],[7,8,9]])
```

```
print numpy.concatenate((array_1, array_2), axis = 1)
```

#Output

```
[[1 2 3 0 0 0]
```

```
[0 0 0 7 8 9]]
```

Task

You are given two integer arrays of size X and X (& are rows, and is the column). Your task is to *concatenate* the arrays along axis .

Input Format

The first line contains space separated integers , and .

The next lines contains the space separated elements of the columns.

After that, the next lines contains the space separated elements of the columns.

Output Format

Print the concatenated array of size X.

Sample Input

4 3 2

1 2

1 2

1 2

1 2

3 4

3 4

3 4

Sample Output

[[1 2]

[1 2]

[1 2]

[1 2]

[3 4]

[3 4]

[3 4]]

HackerRank

PreparePythonNumpyConcatenate

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es

The first line contains space separated integers N , M and P .

The next N lines contains the space separated elements of the P columns.

After that, the next M lines contains the space separated elements of the P columns.

Output Format

Print the concatenated array of size $(N + M) \times P$.

Sample Input

```

4 3 2
1 2
1 2
1 2
1 2
3 4
3 4
3 4

```

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1

Enter your code here. Read input from STDIN. Print output to STDOUT

2

import numpy as np

3

n, m, p = map(int, input().split())

4

a = []

5

b = []

6

for _ in range(n):

7

a.append(list(map(int, input().split())))

8

9

for _ in range(m):

10

b.append(list(map(int, input().split())))

11

12

print(np.concatenate((a, b), axis=0))

Python

★★★★★

You have earned 20.00 points!

You are now 165 points away from the gold level for your python badge.

8%

235/400

Congratulations

You solved this challenge. Would you like to challenge your friends?

f

t

in

Next Challenge

Test case 0

Test case 1

Test case 2

Compiler Message

Success

Input (stdin)

1

4 3 2

2

1 2

3

1 2

4

1 2

5

1 2

6

3 4

7

3 4

8

3 4

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3. [zeros](#)

The `zeros` tool returns a new array with a given shape and type filled with 's.

```
import numpy
```

```
print numpy.zeros((1,2))          #Default type is float
```

```
#Output : [[ 0.  0.]]
```

```
print numpy.zeros((1,2), dtype = numpy.int) #Type changes to int
```

```
#Output : [[0 0]]
```

[ones](#)

The *ones* tool returns a new array with a given shape and type filled with 's'.

```
import numpy
```

```
print numpy.ones((1,2))          #Default type is float
```

```
#Output : [[ 1.  1.]]
```

```
print numpy.ones((1,2), dtype = numpy.int) #Type changes to int
```

```
#Output : [[1 1]]
```

Task

You are given the shape of the array in the form of space-separated integers, each integer representing the size of different dimensions, your task is to print an array of the given shape and integer type using the tools `numpy.zeros` and `numpy.ones`.

Input Format

A single line containing the space-separated integers.

Constraints

Output Format

First, print the array using the `numpy.zeros` tool and then print the array with the `numpy.ones` tool.

Sample Input 0

```
3 3 3
```

Sample Output 0

```
[[[0 0 0]
```

```
 [0 0 0]
```

```
 [0 0 0]]
```

[[0 0 0]

[0 0 0]

[0 0 0]]

[[0 0 0]

[0 0 0]

[0 0 0]]]

[[[1 1 1]

[1 1 1]

[1 1 1]]

[[1 1 1]

[1 1 1]

[1 1 1]]

[[1 1 1]

[1 1 1]

[1 1 1]]]

HackerRank

PreparePythonNumpyZeros and Ones

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Problem

Submissions

Leaderboard

Sample Output 0

[[0 0 0]
[0 0 0]
[0 0 0]

[[0 0 0]
[0 0 0]
[0 0 0]]

[[0 0 0]
[0 0 0]
[0 0 0]]]

[[1 1 1]
[1 1 1]
[1 1 1]]

[[1 1 1]
[1 1 1]
[1 1 1]]]

[[1 1 1]
[1 1 1]
[1 1 1]]]

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1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 import numpy as np
3 shape = tuple(map(int, input().split()))
4 print(np.zeros(shape, dtype=int))
5 print(np.ones(shape, dtype=int))



You have earned 20.00 points!

You are now 145 points away from the gold level for your python badge.



Congratulations

You solved this challenge. Would you like to challenge your friends?

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Test case 0

Compiler Message

Success

Test case 1

Test case 2

Input (stdin)

1 3 3 3

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Test case 3

Expected Output

```
1 [[0 0 0]
2  [0 0 0]
3  [0 0 0]]
```

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